Solutions to

Subtitle or Series

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# Main Title of the Book

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October 2026

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# Chapter 1.

## Basic Problems

**Problem 1. Roots of a quadratic**

Solve the equation $x^2 - 5x + 6 = 0$.

Solution: We look for two numbers whose sum is 5 and whose product is 6: they are 2 and 3, so the polynomial factors as

$$x^2 - 5x + 6 = (x - 2)(x - 3). \tag{1.1}$$

A product is zero only when one of its factors is zero, so $x - 2 = 0$ or $x - 3 = 0$.

**Answer to Problem 1.**

$x = 2$ or $x = 3$.

Verification: Substitution gives $2^2 - 5 \cdot 2 + 6 = 0$ and $3^2 - 5 \cdot 3 + 6 = 0$. The graph in Fig. 1.1 crosses the $x$ axis at these two roots.




Figure 1.1: The polynomial $y = x^2 - 5x + 6$ and its roots.

**Problem 2.**

Let $S_n = 1 + 3 + 5 + \cdots + (2n-1)$ be the sum of the first $n$ odd numbers.

**(a)**

Compute $S_1$, $S_2$, $S_3$ and $S_4$, and guess a formula for $S_n$.

Solution: Adding one odd number at a time,

$$S_1 = 1, \qquad S_2 = 1 + 3 = 4, \qquad S_3 = 4 + 5 = 9, \qquad S_4 = 9 + 7 = 16.$$

These are the squares of 1, 2, 3 and 4.

**Answer to Problem 2 (a).**

$S_n = n^2$.

**(b)**

Prove the formula of part (a) by induction.

Solution: For $n = 1$, $S_1 = 1 = 1^2$. If $S_n = n^2$ for some $n \geq 1$, then

$$S_{n+1} = S_n + (2n+1) = n^2 + 2n + 1 = (n+1)^2, \tag{1.2}$$

so the formula also holds for $n+1$, and by induction for every $n \geq 1$. This part is a proof, so it has no answer box.

**(c) Sum of ten odd numbers**

Evaluate $S_{10}$.

**Answer to Problem 2 (c).**

$S_{10} = 10^2 = 100$ (no solution is needed, so only the answer is given).

**(d)**

How many consecutive odd numbers, starting from 1, must be added for the sum to exceed 1000?

Solution: We need the smallest $n$ with $n^2 > 1000$, that is $n > \sqrt{1000} \approx 31.6$.

**Answer to Problem 2 (d).**

32 odd numbers.

Verification: The preceding sum is $S_{31} = 961$, while $S_{32} = 1024$. Thus 32 is the first value that exceeds 1000.

**Problem 3.**

A right triangle has legs $a = 3$ and $b = 4$, as shown in Fig. 1.2. Find the length of the hypotenuse $c$ and the angle $\theta$ opposite the leg $a$.

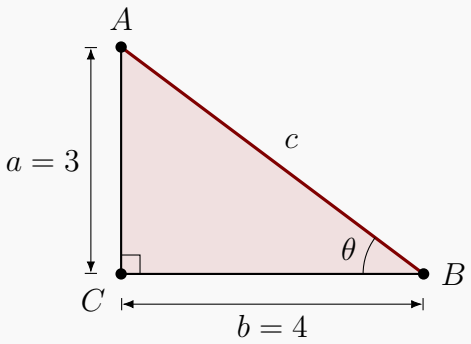



Figure 1.2: Right triangle with legs $a$ and $b$.

Solution: By the Pythagorean theorem,

$$c = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = 5.$$

The leg $a$ is opposite $\theta$ and the leg $b$ is adjacent to it, so

$$\tan\theta = \frac{a}{b} = \frac{3}{4}, \qquad \theta = \arctan\frac{3}{4} \approx 36.87^\circ.$$

**Answer to Problem 3.**

$c = 5$ and $\theta = \arctan(3/4) \approx 36.87^\circ$.

# Chapter 2.

## Figures and References

**Problem 1.**

Consider the function $f(x) = \frac{1}{4}x^2 + \frac{1}{2}$, shown in Fig. 2.1, and the point $P = (3, f(3))$ on its graph.

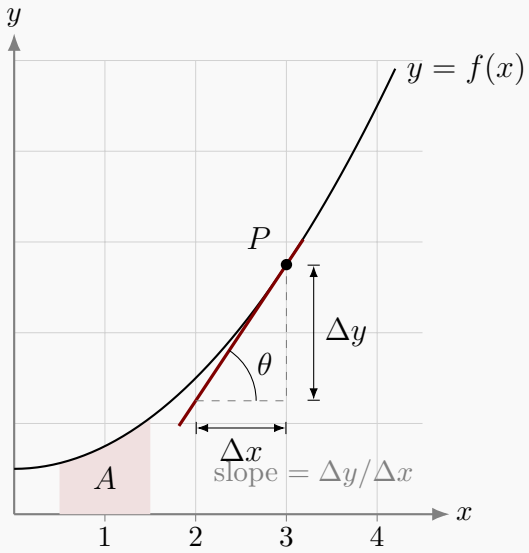



Figure 2.1: Graph of $f$, its tangent at $P$ and the area $A$.

(a)

Find the slope of the tangent at $P$ and the angle $\theta$ that it makes with the $x$ axis.

Solution: The derivative is

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \frac{x}{2},$$

so the slope at $x = 3$ is $f'(3) = 3/2$. In the slope triangle of Fig. 2.1, $\Delta y/\Delta x = \tan\theta$, hence $\theta = \arctan(3/2)$.

**Answer to Problem 1 (a).**

Slope $3/2$, and $\theta = \arctan(3/2) \approx 56.31°$.

(b)

Find the equation of the tangent at $P$.

Solution: Since $f(3) = \frac{9}{4} + \frac{1}{2} = \frac{11}{4}$, the line through $P$ with slope $3/2$ is

$$y = \frac{11}{4} + \frac{3}{2}(x - 3) = \frac{3}{2}x - \frac{7}{4}.$$

**Answer to Problem 1 (b).**

$y = \frac{3}{2}x - \frac{7}{4}$.

(c)

Find the area $A$ under the graph of $f$ between $x = \frac{1}{2}$ and $x = \frac{3}{2}$.

Solution:

$$A = \int_{1/2}^{3/2} \left( \frac{x^2}{4} + \frac{1}{2} \right) \mathrm{d}x = \left[ \frac{x^3}{12} + \frac{x}{2} \right]_{1/2}^{3/2} = \frac{27-1}{96} + \frac{1}{2} = \frac{13}{48} + \frac{24}{48}.$$

**Answer to Problem 1 (c).**

$A = \frac{37}{48} \approx 0.771.$

**Problem 2.**

Use Euclid's algorithm to find the greatest common divisor of 84 and 36.

Solution:

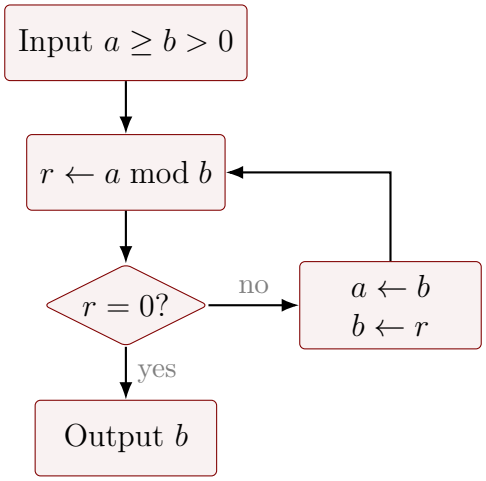



Figure 2.2: Euclid's algorithm.

Starting from $a = 84$ and $b = 36$, each step of the algorithm (Fig. 2.2) replaces $(a, b)$ by $(b, a \bmod b)$:

| Step | $a$ | $b$ | $r = a \bmod b$ |
|---|---|---|---|
| 1 | 84 | 36 | 12 |
| 2 | 36 | 12 | 0 |

The remainder is zero at step 2, so the algorithm outputs $b = 12$. Indeed $84 = 7 \cdot 12$ and $36 = 3 \cdot 12$, with $\gcd(7, 3) = 1$.

**Answer to Problem 2.**

$\gcd(84, 36) = 12$.

**Problem 3.**

For which $n$ does the sum $S_n$ of Problem 1.2 satisfy $S_n = 5n - 6$?

Solution: By part 1.2(a) (proved in Chap. 1 Prob. 2), $S_n = n^2$, so the condition becomes $n^2 - 5n + 6 = 0$. This is the equation of Chap. 1 Prob. 1, whose factorization (1.1) gives $n = 2$ or $n = 3$. Both are positive integers, and indeed $S_2 = 4 = 5 \cdot 2 - 6$ and $S_3 = 9 = 5 \cdot 3 - 6$.

**Answer to Problem 3.**

$n = 2$ or $n = 3$.

**Problem 4.** *Not done*

Show that the sum of the first $n$ even numbers, $2+4+\cdots+2n$, is $n(n+1)$. (Hint: compare it with $S_n$, or use induction as in Problem 1.2.)

**Problem 5.**

Use the two panels in Fig. 2.3 to answer the following questions.

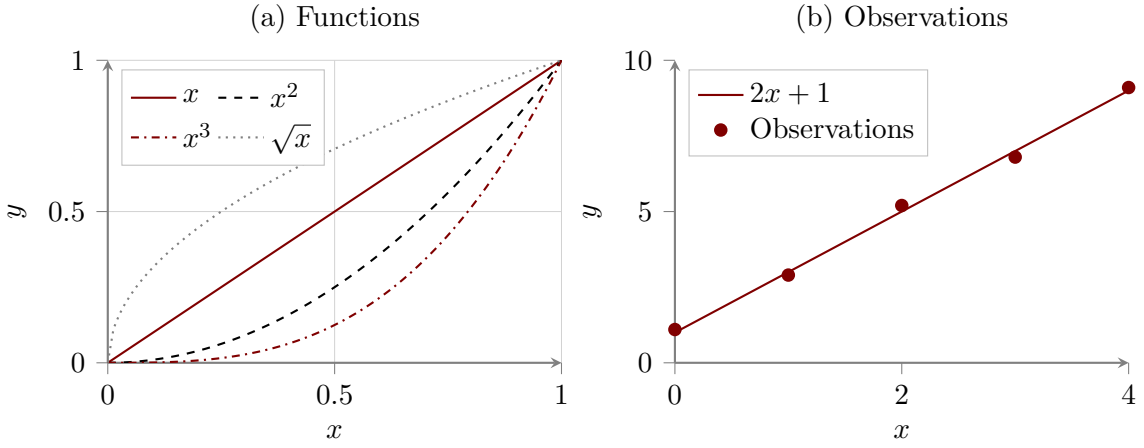



Figure 2.3: Functions on the unit interval and observations of $y = 2x + 1$.

(a)

Order $x$, $x^2$, $x^3$ and $\sqrt{x}$ for $0 < x < 1$.

Solution: Multiplication by $0 < x < 1$ decreases a positive number, so $x^3 < x^2 < x$. Also $x^2 < x$ implies $x < \sqrt{x}$.

**Answer to Problem 5 (a).**

$x^3 < x^2 < x < \sqrt{x}$.

(b)

What is the largest absolute deviation of the five observations from $y = 2x + 1$?

Solution: The absolute deviations $|y_{\text{measured}} - (2x + 1)|$ at $x = 0, 1, 2, 3, 4$ are 0.1, 0.1, 0.2, 0.2 and 0.1, respectively.

**Answer to Problem 5 (b).**

The largest absolute deviation is 0.2, at $x = 2$ and $x = 3$.

# Chapter 3.

## Sections and Progress

## 3.1 Sequences

**Problem 1.**

Compute the sum $1 + 2 + 3 + \cdots + 100$.

Solution: Pair the first term with the last, the second with the second to last, and so on: each of the 50 pairs $k + (101 - k)$ adds up to 101.

**Answer to Problem 1.**

$1 + 2 + \cdots + 100 = 50 \cdot 101 = 5050$.

**Problem 2.**

An arithmetic sequence starts with $3, 7, 11, \ldots$. Find its tenth term $a_{10}$ and the sum of its first ten terms.

Solution: The common difference is 4, so $a_n = 3 + 4(n-1)$ and $a_{10} = 39$. The sum of an arithmetic sequence is the number of terms times the mean of the first and last terms:

$$a_1 + \cdots + a_n = \frac{n(a_1 + a_n)}{2}. \tag{3.1}$$

**Answer to Problem 2.**

$a_{10} = 39$, and the sum is $\frac{10(3+39)}{2} = 210$.

**Problem 3.** *Not done*

Find a closed form for the sum of the first $n$ squares, $1^2 + 2^2 + \cdots + n^2$.

**Problem 4.** *Not done*

Show that the sequence $a_n = \left(1 + \frac{1}{n}\right)^n$ is increasing and bounded above by 3.

Solution: By the binomial theorem,

$$a_n = \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k} = \sum_{k=0}^{n} \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right).$$

Each product grows with $n$, and $a_{n+1}$ has one more positive term than $a_n$, so the sequence is increasing. The bound remains to be proved.

## §12 Series

**Problem 1.**

Find the sum of the geometric series $1 + \frac{1}{2} + \frac{1}{4} + \dots$.

Solution: The partial sums are

$$S_n = \sum_{k=0}^{n-1} \frac{1}{2^k} = \frac{1 - 2^{-n}}{1 - \frac{1}{2}} = 2\left(1 - 2^{-n}\right),$$

and $2^{-n} \to 0$.

**Answer to Problem 1.**

The sum is 2.

**Problem 2.** *Not done*

Consider the series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$.

(a)

Write $\frac{1}{k(k+1)}$ as a difference of two simple fractions.

Solution: Since $(k+1)-k=1$,

$$\frac{1}{k}-\frac{1}{k+1}=\frac{(k+1)-k}{k(k+1)}=\frac{1}{k(k+1)}.$$

**Answer to Problem 2 (a).**

$\frac{1}{k(k+1)}=\frac{1}{k}-\frac{1}{k+1}$.

(b) *Not done*

Compute the partial sum $S_n=\sum_{k=1}^{n}\frac{1}{k(k+1)}$.

(c)

Deduce the sum of the series.

**Answer to Problem 2 (c).**

The sum is 1.

**Problem 3.**

Show that the series $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges.

Solution: For $k \geq 2$, $\frac{1}{k^2} < \frac{1}{(k-1)k}$. By part §12.2(a) (§12 Prob. 2), these terms telescope:

$$\sum_{k=1}^{n} \frac{1}{k^2} < 1 + \sum_{k=2}^{n} \left( \frac{1}{k-1} - \frac{1}{k} \right) = 2 - \frac{1}{n} < 2.$$

The partial sums increase and are bounded, so the series converges.

**Answer to Problem 3.**

The series converges, and its sum is less than 2.

**Problem 4.**

Find the sum of the first 20 terms of the sequence of Problem 3.1.2, and compare it with the sum of the first 20 odd numbers.

Solution: By (3.1) (Sec. 3.1 Prob. 2), with $a_{20} = 3 + 4 \cdot 19 = 79$, the sum is $\frac{20(3+79)}{2} = 820$. The sum of the first 20 odd numbers is $20^2 = 400$ (Chap. 1 Prob. 2), and the series of Problem §12.1 shows that sums of infinitely many terms can still be finite.

**Answer to Problem 4.**

820, against 400 for the odd numbers.

# Chapter 4.

## Showcase

## Problem 1. Sectors and areas *Not done*

Fig. 4.1 shows the point $P$ at the angle $\theta$ and the area $A$.

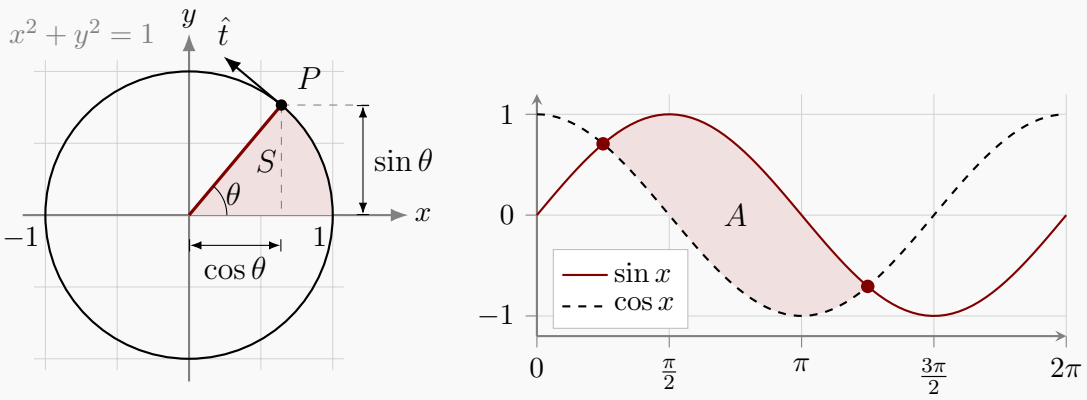



Figure 4.1: Left: the sector $S$ of angle $\theta$. Right: the area $A$ between $\sin x$ and $\cos x$.

### (a) Area of a sector

Find the area of the sector $S$.

Solution: The sector is the fraction $\theta/2\pi$ of the disk, of area $\pi r^2$:

$$S = \frac{\theta}{2\pi}\,\pi r^2 = \frac{\theta r^2}{2}. \tag{4.1}$$

### Answer to Problem 1 (a).

$S = \theta/2$ for $r = 1$.

### (b)

Find the area $A$.

Solution: The curves cross at $x = \pi/4$ and $5\pi/4$. As in Chap. 2 Prob. 1,

$$A = \int_{\pi/4}^{5\pi/4} (\sin x - \cos x)\,dx = \Big[-\cos x - \sin x\Big]_{\pi/4}^{5\pi/4} = \sqrt{2} + \sqrt{2}.$$

### Answer to Problem 1 (b).

$A = 2\sqrt{2} \approx 2.83$.

Verification: As $\sin x - \cos x = \sqrt{2}\sin(x - \pi/4)$, $A$ is $\sqrt{2}$ times one arch of sin, of area 2: the sector (4.1) of part 4.1(a) with $\theta = 4\sqrt{2}$.

(c) *Not done*

Show that $\sin^2\theta + \cos^2\theta = 1$ from the lengths in Fig. 4.1.